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/ **Practice Exam - 1**

Correct Answer

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Practice Exam - 1

Question 15 of 117



A 72-year-old woman was referred for evaluation of progressive dyspnea and fatigue. You are the third cardiologist she sees because the etiology of her symptoms remains elusive. She has a long-standing hypertension, dyslipidemia, T2DM, carpal tunnel syndrome. She can no longer climb a flight of stairs without stopping frequently due to gradual worsening of dyspnea on exertion over the last 6 months. She denies orthopnea, leg edema, chest pain or dizziness. Her current medications include Losartan 50 mg/d, amlodipine 5mg/d, atorvastatin 20 mg/d, and metformin 500 mg twice a day.

A previous echocardiogram showed normal LV and RV systolic function, mildly biatrial enlargement, and no valvular pathology. Systolic RV pressure was 34 mmHg and E/E' ratio was 8.5 by doppler. Pulmonary function tests were normal. CXR was unremarkable. Coronary arteries were normal angiographically with LVEDP of 12mmHg.

Physical exam revealed an oriented patient with BP 127/80 mmHg and HR 78bpm, BMI 28 kg/m², no neck vein distension, clear lungs, regular rate with no cardiac gallop and no leg edema.

Laboratory data: A1C 6.7%; NT-pro-BNP 390 pg/ml; creatinine 0.9 mg/dl; potassium 4.3 mEq/L, sodium 139 mEq/L.

Which of the following tests is most likely to uncover the diagnosis of the patient's exertional dyspnea?

- A. Chest computed tomography with PE protocol.
-  B. Right heart catheterization with 300 ml normal saline volume challenge.
- C. High resolution chest computed tomography.
- D. Right ventricular endomyocardial biopsy.
-  E. Right heart catheterization with exercise hemodynamics.

Testing Point: The AHFTC should know that resting hemodynamics are inadequate to diagnose exertional symptoms and exercise hemodynamics are required in this setting.

Answer: E. Right heart catheterization with exercise hemodynamics. Measurement of hemodynamics during exercise can be very helpful to determine the cause of exertional dyspnea in patients with unremarkable resting cardiac work-up and pulmonary work-up. Normal PCWP and normal NT-pro-BNP at rest do not preclude potential presence of a cardiac problem. Increase in PCWP above 25 mmHg with exercise suggests cardiac etiology of dyspnea on exertion, consistent most likely with undiagnosed HFpEF. Provocative RHC can be very useful in these patients. Abnormal rise in pulmonary systolic pressure above 45 mmHg with abnormal rise in PCWP above 25 mmHg during exercise is highly diagnostic to identify HFpEF with very high sensitivity 96% and specificity 95%. Normal saline bolus (300ml) during RHC may uncover the diagnosis, but may not be sufficient and exercise hemodynamics are more reliable to diagnose HFpEF in cath lab.

Thus E is a better choice than B.

Answer A: Normal RV and only mildly elevated RVSP with no TR speak against significant pulmonary emboli.

Answer C: Chest HRCT is unlikely to be positive for pulmonary parenchymal abnormalities with normal PFTs and normal CXR.

Answer D: Heart biopsy is not indicated without suggestion of an inflammatory or infiltrative process in myocardium. Exercise hemodynamics have been shown to enhance the diagnosis of unexplained exertional dyspnea. Distractors: carpal tunnel syndrome, mild biatrial enlargement.

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